Example 2.22

Find the modulus and principal argument of the following complex numbers.

(i) $\sqrt{3}+i$ (ii) $-\sqrt{3}+i$ (iii) $-\sqrt{3}-i$ (iv) $\sqrt{3}-i$

Solution

(i) $\sqrt{3}+i$

$$\text{Modulus} = \sqrt{x^2+y^2} = \sqrt{\left(\sqrt{3}\right)^2+1^2} = \sqrt{3+1} = 2$$

$$\alpha = \tan^{-1}\left|\frac{y}{x}\right| = \tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{6}$$

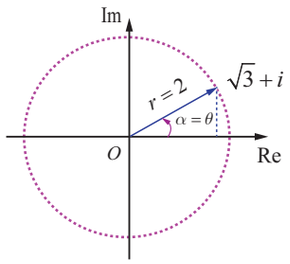


Fig. 2.36

Since the complex number $\sqrt{3}+i$ lies in the first quadrant, has the principal value

$$\theta = \alpha = \frac{\pi}{6}.$$

Therefore, the modulus and principal argument of $\sqrt{3}+i$ are 2 and $\frac{\pi}{6}$ respectively.

(ii) $-\sqrt{3}+i$

Modulus = 2 and

$$\alpha = \tan^{-1}\left|\frac{y}{x}\right| = \tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{6}$$

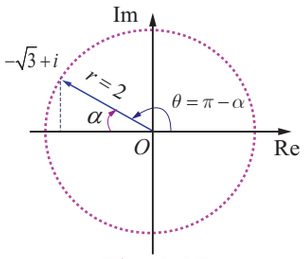


Fig. 2.37

Since the complex number $-\sqrt{3}+i$ lies in the second quadrant has the principal value

$$\theta = \pi - \alpha = \pi - \frac{\pi}{6} = \frac{5\pi}{6}.$$

Therefore the modulus and principal argument of $-\sqrt{3}+i$ are 2 and $\frac{5\pi}{6}$ respectively.

(iii) $-\sqrt{3}-i$

$$r = 2 \text{ and } \alpha = \frac{\pi}{6}.$$

Since the complex number $-\sqrt{3}-i$ lies in the third quadrant, has the principal value,

$$\theta = \alpha - \pi = \frac{\pi}{6} - \pi = -\frac{5\pi}{6}.$$

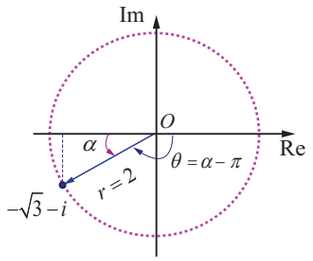


Fig. 2.38

Therefore, the modulus and principal argument of $-\sqrt{3}-i$ are 2 and $-\frac{5\pi}{6}$ respectively.

(iv) $\sqrt{3}-i$

$$r = 2 \text{ and } \alpha = \frac{\pi}{6}.$$

Since the complex number lies in the fourth quadrant, has the principal value,

$$\theta = -\alpha = -\frac{\pi}{6}$$

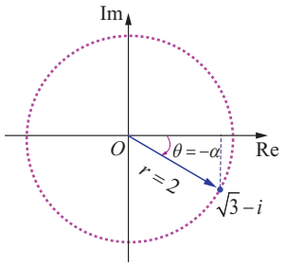


Fig. 2.39

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